Ben-Chi Ma | Curriculum Vitae

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Department of Aerospace and Mechanical Engineering

EDUCATION

University of Notre Dame — **Notre Dame, IN, USA**
Ph.D. in Mechanical Engineering — *Aug. 2024–Present*
Department of Aerospace and Mechanical Engineering, Advisor: Prof. Tengfei Luo
Research focus: AI-accelerated multiscale materials modeling and phonon thermal transport in semiconductors and at interfaces

Xi'an Jiaotong University — **Xi'an, Shaanxi, China**
M.Eng. (research degree, thesis), GPA: 3.64/4.0 (Rank 2/105) — *Sept. 2021–Jun. 2024*
Major in Power Engineering and Engineering Thermophysics, School of Energy and Power Engineering
Thesis topic: Optical regulation of nanomaterials for spectral splitting and passive radiative sub-ambient cooling application in solar cell
• ***Awards: National Scholarship, Excellent Postgraduate Student Cadre, Qiyiyi Research Institution Scholarship, Outstanding Volunteer in 11th National Paralympic Games and the 8th Special Olympics (Xi'an, Shaanxi)***

Xi'an Jiaotong University — **Xi'an, Shaanxi, China**
B.Eng. in Mechanical Engineering, GPA: 3.80/4.0 (Rank 1/31) — *Sept. 2017–Jul. 2021*
Major in New Energy Science and Engineering, School of Energy and Power Engineering
• ***Awards: The First Prize Scholarship, The Second Prize Scholarship, YINGDE Gases Scholarship, Chen Xuejun Undergraduate Scholarship, Excellent Student, Excellent Student Cadre, Outstanding League Cadres, First Prizes of Shaanxi province of 2018 and 2019 China Undergraduate Mathematical Contest in Modeling (CUMCM), Meritorious Winner in Mathematical Contest in Modeling (MCM), No.1 in National 100 Excellent Graduation Theses (Design) Competition in Energy and Power.***
• Attended the 2019 Hong Kong Polytechnic University Summer International School and studied two courses gaining A and A- grades respectively.

PROFESSIONAL EXPERIENCE

Doctoral Research at University of Notre Dame

Graduate Research Assistant — **Notre Dame, IN, USA**
Advisor: Prof. Tengfei Luo — *Aug. 2024–Present*

- **Phonon spectra and interfacial thermal transport at GaN/Al_2O_3 dislocation cores.** Ran molecular dynamics with a foundation machine-learning interatomic potential (MACE) on ∼1200-atom interface models. Resolved region-specific phonon density of states and EELS-like spectra at dislocation core and bridge sites, and evaluated elastic interfacial thermal conductance with the atomistic Green's function method (manuscript in preparation).
- **Machine-learning potentials for thermal conductivity of InGaAs.** Benchmarking self-trained Deep Potential models against foundation MLIPs (MACE, DPA) for phonon dispersion and lattice thermal conductivity of $In_{1-x}Ga_xAs$, using phono3py and the virtual-crystal approximation; building a reproducible MLIP-to-phonon workflow.
- **Multiscale thermal modeling of GaN power and RF transistors** (sponsored by Lockheed Martin). Coupled GPU-accelerated multi-band phonon Boltzmann transport simulations with Fourier conduction to resolve non-Fourier near-junction heating, and quantified junction-to-backside thermal resistance of GaN-on-Si power FETs versus substrate thinning and carrier choice. Contributed to physics-informed neural network solvers for the phonon BTE.
- **Phonon band discretization for the deterministic BTE solver JAX-BTE.** Developed frequency–mean-free-path band binning with κ-conserving band aggregation, replacing the original row-order binning: among the schemes benchmarked, it is the only one that keeps both bulk thermal transport and interface transmission accurate at a fixed band budget. Added a solver-free rule for splitting the band budget between frequency windows and MFP sub-bins, and benchmarked bulk, thin-film, interface and TDTR cases against a reference solver (manuscript in preparation).

Course Projects — **Notre Dame, IN, USA**
CSE 60772 High Performance Distributed Systems — *Fall 2026*

- **GPU (CUDA, C):** ported a 1D phonon BTE solver (up to 16.8 million position–band–direction states), a heat-diffusion stencil and a Mandelbrot kernel to CUDA; GPU-resident data and CUDA Graphs gave 146–163× (BTE) and up to 599× speedups over one CPU core on A100 GPUs, with checksums matching the CPU code on 5 GPU types.
- **Multicore and distributed:** OpenMP tasking and SIMD reached 42× (Mandelbrot) and 38× (N-body) on 64 AMD EPYC cores; AVX2 intrinsics with masked blends ran 3–4× faster than scalar code; an HTCondor workflow rendered 1,440 frames on 70 hosts (peak 1,204 cores) for a 122× wall-clock speedup.
- **Genomics pipeline:** accelerated a *P. falciparum* whole-genome variant-calling pipeline (BWA-MEM, Picard, GATK) for bulk segregant analysis 5.77× end to end (5,686 s → 985 s on 16 cores) without modifying the tools, through chromosome-level scatter-gather of the single-threaded GATK stages and a streaming align/sort/deduplicate pipe; results verified against the baseline (99.6% site concordance). Now extending it to distributed execution with TaskVine on HTCondor.

Teaching at University of Notre Dame

Teaching Assistant — **Notre Dame, IN, USA**
AME 21268 Design Tools II (undergraduate) — *Fall 2024–Present*

- In-class teaching assistant; grade homework and final assessments.
- Guide students in SolidWorks 3D CAD modeling, CAM simulation, engineering mechanics and structural analysis, Ackermann steering geometry, and engineering design projects.

Guest Instructional Session — **Notre Dame, IN, USA**
Academic English Writing (non-credit course) — *Fall 2024*

- Introduced Zotero for academic writing: literature organization, citation and bibliography management, and writing workflows.

Research in State Key Laboratory of Multiphase Flow in Power Engineering

Graduate Researcher — **Xi'an, China**
MFREC Group, supervisor: Prof. Dengwei Jing (dwjing@mail.xjtu.edu.cn) — *Sept. 2021–Jun. 2024*

- Focused on the optical regulation of nanomaterials for spectral splitting, solar energy conversion in CPV/T system and passive radiative sub-ambient cooling coatings application in solar cell. Co-authored **11** SCI papers (including **2** as first author), **1** invention patent and **1** software copyright.
- Made contributions to 5 national programs and foundations in our Lab.
- Had experiences in setup of CPVT-Hydrogen system, preparation and characterization of nanomaterials and polymer-dielectric composite materials, and testing of thermal and optical properties, numerical simulation through COMSOL Multiphysics.

Teaching Assistant — **Xi'an, China**
Undergraduate Course: Hydrogen and New Energy Power Systems — *Sept. 2021–Dec. 2022*

- Taught laboratory courses for **82** students up to **8** hours, evaluating and totaling student grades.

Key R&D projects with corporate cooperation

Group Member — **Xi'an, China & Tacheng, China**
Solar Photo/ Thermal/ Electric Coupling Hydrogen Production Project — *Sept. 2021–Sept. 2022*

- Cooperated with State Power Investment Group Xinjiang Energy Chemical Co., Ltd. to complete the industrialization demonstration project of solar photo/thermal/electric coupling hydrogen production technology.
- Worked together with the team to iterate and upgrade the system, and have built 50 system setups in Tacheng, Xinjiang.

Undergraduate Research Project Design in Xi'an Jiaotong University

Group Leader — **Xi'an, China & Tacheng, China**
Fully self-sustaining solar absorption based water harvesting unit — *Jan. 2020–Sept. 2021*

- Led a group of **5** undergraduates and finished the project. We received an A+ grade in program evaluation, published **2** related review articles and won **5** science and innovation competitions. I optimized the design of the heat transfer structure by numerical simulation, built the device and tested the performance in the team.

PUBLICATIONS

Journal Articles

[1] Jiahang Zhou, Ruiyang Li, Wenjie Shang, Yi Liu, J.P. Panda, Pan Du, Xin-Yang Liu, Jasmine Liang, **Ben-Chi Ma**, Jian-Xun Wang, and Tengfei Luo. Physics-informed neural networks with hard-encoded angle-dependent boundary conditions for phonon Boltzmann transport equation. *Materials Today Physics*, 59:101922, 2025. https://doi.org/10.1016/j.mtphys.2025.101922

[2] Qingxi Xin, **Benchi Ma**, Jiaying Ru, Yu Zhou, and Dengwei Jing. Efficient passive cooling over a novel bifunctional polymer bilayer composite simultaneously possessing radiative and evaporative cooling properties. *Advanced Energy Materials*, 15(14):2404122, 2025. https://doi.org/10.1002/aenm.202404122

[3] Jiwei Li, **Benchi Ma**, Hongxia Wang, Yizhou Zhu, Yujie Gu, and Dengwei Jing. A novel photovoltaic-thermal system based on spectral splitting of nanoparticle suspensions for simultaneous hydrogen production and seawater desalination. *Energy Conversion and Management*, 314:118670, 2024. https://doi.org/10.1016/j.enconman.2024.118670

[4] **Ben-Chi Ma**, Maochang Liu, Yizhou Zhu, Kefan Chen, Lixia Pan, Chaoqian Ai, and Dengwei Jing. Record passive radiative cooling of over 15 ℃ by a micro-nanoporous granular composite film. *ACS Applied Energy Materials*, 7(12):5064–5070, 2024. https://doi.org/10.1021/acsaem.4c00675

[5] Xinlong Lu, Devendra Yadav, **Benchi Ma**, Lijing Ma, and Dengwei Jing. Rapid detachment of hydrogen bubbles for electrolytic water splitting driven by combined effects of Marangoni force and the electrostatic repulsion. *Journal of Power Sources*, 599:234217, 2024. https://doi.org/10.1016/j.jpowsour.2024.234217

[6] Devendra Yadav, Xinlong Lu, **Ben-Chi Ma**, and Dengwei Jing. Advancements in microreactor technology for hydrogen production via steam reforming: A comprehensive review of numerical studies. *Journal of Power Sources*, 596:234090, 2024. https://doi.org/10.1016/j.jpowsour.2024.234090

[7] Yizhou Zhu, **Benchi Ma**, Baichuan He, Xinyu Ma, and Dengwei Jing. Liquid spherical lens as an effective auxiliary optical unit for CPV/T system with remarkable hydrogen production efficiency. *Applied Energy*, 334:120733, 2023. https://doi.org/10.1016/j.

apenergy.2023.120733

[8] **Ben-Chi Ma**, Hua Lin, Yizhou Zhu, Zilong Zeng, Jiafeng Geng, and Dengwei Jing. A new Concentrated Photovoltaic Thermal-Hydrogen system with photocatalyst suspension as optical liquid filter. *Renewable Energy*, 194:1221–1232, 2022. https://doi.org/10.1016/j.renene.2022.06.013

[9] Yizhou Zhu, **Benchi Ma**, Zilong Zeng, Hewei Lou, Yi He, and Dengwei Jing. Solar collector tube as secondary concentrator for significantly enhanced optical performance of LCPV/T system. *Renewable Energy*, 193:418–433, 2022. https://doi.org/10.1016/j.renene.2022.04.165

[10] Zhihui Chen, Jinwen Shi, Yueqi Li, **Benchi Ma**, Xueli Yan, Maochang Liu, Hui Jin, Dong Li, Dengwei Jing, and Liejin Guo. Recent progress of energy harvesting and conversion coupled with atmospheric water gathering. *Energy Conversion and Management*, 246:114668, 2021. https://doi.org/10.1016/j.enconman.2021.114668

[11] Zhihui Chen†, Shiyu Song†, **Benchi Ma**†, Yueqi Li, Yu Shao, Jinwen Shi, Maochang Liu, Hui Jin, and Dengwei Jing. Recent progress on sorption/desorption-based atmospheric water harvesting powered by solar energy. *Solar Energy Materials and Solar Cells*, 230:111233, 2021. https://doi.org/10.1016/j.solmat.2021.111233

[12] Zilong Zeng, Lijun Sun, Heng Liu, **Benchi Ma**, Dengwei Jing, and Liejin Guo. Should the tubular photocatalytic reactors work continuously or in an intermittent manner instead? *Industrial & Engineering Chemistry Research*, 60(12):4610–4621, 2021. https://doi.org/10.1021/acs.iecr.1c00463

† Equal contribution. Full record: ORCID 0000-0002-1556-5974 · Google Scholar

CONFERENCE PRESENTATIONS

SciML Workshop at the University of Notre Dame (SciML@ND 2026) — **Notre Dame, IN, USA**
Poster: Machine-learning-assisted multiscale phonon transport in AlGaN/GaN HEMTs — *Sept. 2026*

42nd International Conference on Thermoelectrics (ICT 2026) — **Houston, TX, USA**
Oral: Concurrent-learning deep potential for thermal transport in $In_{1-x}Ga_xAs$ alloys (B.-C. Ma, Z. Xu, T. Luo) — *May 2026*

ASME 2025 Summer Heat Transfer Conference (SHTC 2025) — **Westminster, CO, USA**
Technical presentation (SHTC2025-164577) — *Jul. 2025*
Multiscale thermal transport modeling for GaN transistors: Bridging first-principles and the phonon Boltzmann equation

Annual Conference on Multiphase Flow of Chinese Society of Engineering Thermophysics — **Suzhou, China**
Study on the compatibility of spectral selective regulation between Pt@TiO_2 photothermal — *Dec. 2021*

Annual Conference on Multiphase Flow of Chinese Society of Engineering Thermophysics — **Shanghai, China**
Spectrally Modulated Thermoplastic Polyurethane Films with Gas-Solid Micro/Nanostructures for Radiative Cooling — *Oct. 2023*

INTERNSHIPS

Suzhou Transportation Investment Group Co., Ltd. — **Suzhou, China**
Intern — *Jul. 2020–Aug. 2020*

- Completed a research report on the state of the art and development of the hydrogen industry, which was appreciated by the Chairman of the branch, as well as won the honors of Outstanding Research Report and Outstanding Internship in the competition.

SKILLS

Computational

Materials modeling: Machine-learning interatomic potentials (MACE, DPA, DeePMD), molecular dynamics (ASE), lattice dynamics and phonon transport (phono3py, ALAMODE), deterministic phonon BTE (JAX-BTE), atomistic Green's function, finite-volume and multiscale BTE–Fourier thermal modeling

Programming: Python (JAX, NumPy/SciPy), Git/GitHub, Bash, LaTeX; experience with C++, Fortran and MATLAB

HPC: C, CUDA (CUDA Graphs, streams, pinned memory), OpenMP (tasking, SIMD), AVX2 intrinsics; GPUs (A100, V100, A10, RTX 6000); Slurm, HTCondor and UGE clusters; scatter-gather decomposition, streaming pipelines and distributed task graphs (TaskVine); profiling and strong/size scaling analysis (Amdahl's law); reproducible pipelines with NAS backup

AI-assisted research: AI-assisted software development with coding agents (Claude Code, Codex); designed a multi-agent research workflow with a Git-versioned shared project memory and review-before-merge governance across multiple machines and clusters

Engineering: COMSOL Multiphysics, SolidWorks, Inventor, CAD, Matlab/Simulink, NI LabVIEW

Experiment

- Skilled in material preparation, system design, and characterization like GC, Spectrophotometry, SEM, XRD, FT-IR, etc.

Languages

Chinese: Native Language

English: TOEFL iBT — *Scores:91*